

Short-Term Object Interaction Anticipation – Benchmark Results Summary

This document summarizes reported results from several works related to Ego4D Short-Term Object Interaction Anticipation (STA) and related benchmarks. For each paper, we include the main quantitative tables and a short explanation of what each table shows.

1. AFF-ttention! Affordances and Attention Models for Short-Term Object Interaction Anticipation

1.1 Ego4D-STA v2 – Validation Split

Results in Top-5 mAP (%) on the validation split of Ego4D-STA v2. Higher is better. **N** = noun-only, **N+V** = noun + verb, **N+ δ** = noun + TTC-bin, **All** = noun + verb + TTC-bin.

Model	N	N+V	N+ δ	All
FRCNN+SF [17]	21.00	7.45	7.07	2.98
InternVideo [4]	19.45	8.00	6.97	3.25
StillFast [42]	20.26	10.37	7.26	3.96
GANO v2 [50]	20.52	10.42	7.28	3.99
STAformer	24.85	13.45	7.41	4.90
STAformer + AFF	27.03	14.36	8.72	5.04
STAformer + MH	27.51	14.68	9.63	5.50
STAformer + MH + AFF	29.39	15.38	9.94	5.67
Gain (rel %) vs 2nd best	+43.3	+47.6	+36.5	+42.1

Explanation.

This table compares a range of baselines (FRCNN+SF, InternVideo, StillFast, GANO v2) against STAformer and its variants on Ego4D-STA v2 validation. Adding multi-head attention (MH) and the proposed affordance attention (AFF) on top of STAformer yields the best performance across all semantic metrics (N, N+V, N+ δ , All), with a large relative gain over the second-best model.

1.2 EPIC-Kitchens – Validation Split

Results in Top-5 mAP (%) on the validation split of EPIC-Kitchens.

Model	N	N+V	N+ δ	All
StillFast [42]	21.24	12.41	6.22	3.28

Model	N	N+V	N+ δ	All
STAformer	24.16	15.55	7.08	4.31
STAformer + AFF	26.19	16.49	7.18	4.69
Gain (rel %) vs 2nd best	+23.3	+32.8	+15.4	+42.9

Explanation.

On EPIC-Kitchens, STAformer already improves over StillFast in all metrics. Incorporating affordance attention (AFF) further boosts performance, especially on joint metrics such as N+V and All. The relative gain row shows the percentage improvement of STAformer+AFF compared to the second-best model (plain STAformer).

2. Short-Term Object Interaction Anticipation with Disentangled Object Detection (SOIA-DOD)

2.1 Ablation: Number of Object Candidates

Top-5 mAP (%) on Ego4D validation for different numbers of object candidates per frame used as transformer queries.

# Candidates	Noun (N)	N+V	N+TTC	Overall
5	30.14	14.54	9.219	4.91
10	30.65	15.22	9.222	4.98
20	30.94	14.88	8.86	4.87

Explanation.

This ablation studies how many potential active object candidates are passed to the transformer. Increasing from 5 to 10 candidates slightly improves most metrics (especially N+V and Overall). Pushing to 20 candidates gives a marginal gain on noun-only mAP but degrades N+TTC and Overall, suggesting that too many noisy proposals can hurt anticipation quality.

2.2 Comparison with Baselines on Ego4D v2 Test

Top-5 mAP (%) on the Ego4D v2 **test** set.

Method	Noun (N)	N+V	N+TTC	Overall
StillFast [6]	25.06	13.29	9.14	5.12
GANov2 [7]	25.67	13.60	9.02	5.16
Zarrio*	33.50	17.26	11.77	6.75
EgoVideo*	31.08	16.18	12.41	7.21
SOIA-DOD (ours)	34.89	17.61	10.91	6.22

Explanation.

This table evaluates multiple anticipation methods on the official Ego4D test set. Methods marked with * (Zarrio, EgoVideo) use stronger video backbones. SOIA-DOD achieves the best noun and N+V scores and competitive N+TTC and Overall performance, showing that disentangling object detection from anticipation helps improve short-term interaction prediction.

3. Egocentric Object-Interaction Anticipation with Retentive and Predictive Learning (EgoIR / EgoIP)

Results in Top-5 mAP (%) on Ego4D-STA v2. Columns correspond to different label combinations: **b+n** (binary + noun), **b+v** (binary + verb), **b+n+t** (binary + noun + TTC), **b+n+v** (binary + noun + verb), **b+n+v+t** (binary + noun + verb + TTC).

Model (mAP)	b+n	b+v	b+n+t	b+n+v	b+n+v+t
One-stage training					
StillFast	20.26	–	7.16	10.37	3.96
STAformer	24.85	–	7.41	13.45	4.90
StillFast + D EgoIR	25.75	12.35	9.30	14.41	5.56
StillFast + D EgoIP	26.00	11.92	9.62	14.44	5.74
Two-stage training					
SlowFast	21.00	–	7.04	7.45	2.98
EgoAnticipator†	22.54	10.89	9.67	10.28	5.41
EgoAnticipator	23.52	11.18	10.29	11.04	5.60

† uses teacher–student configuration with $\phi_M = \phi_S$.

Explanation.

This table compares baseline models (StillFast, STAformer, SlowFast) with the proposed retentive/predictive variants EgoIR and EgoIP, as well as EgoAnticipator. Adding the retentive–predictive learning (D EgoIR / D EgoIP) on top of StillFast improves all metrics, especially those that combine binary and semantic supervision (b+n+t, b+n+v, b+n+v+t). Two-stage training with EgoAnticipator further refines performance compared to SlowFast.

4. StillFast: An End-to-End Approach for Short-Term Object Interaction Anticipation

Results in Top-5 mAP (%) on Ego4D v2 **validation** and **test** splits.

Split	Method	Ver	Noun (N)	N+V	N+TTC	Overall
Val	FRCNN+SF [17]	v2	21.00	7.45	7.04	2.98
Val	StillFast (ours)	v2	20.26	10.37	7.16	3.96

Split	Method	Ver	Noun (N)	N+V	N+TTC	Overall
Test	FRCNN+SF [17]	v2	26.15	9.45	8.69	3.61
Test	StillFast (ours)	v2	25.06	13.29	9.14	5.12

Explanation.
The table compares the original FRCNN+SF pipeline with the end-to-end StillFast model. On both validation and test splits, StillFast trades a tiny drop in noun-only accuracy for substantial gains on joint metrics (N+V, N+TTC, Overall), demonstrating the benefit of learning detection and anticipation jointly in a single network.

5. Guided Attention for Next Active Object (GANOV2) – Ego4D STA Challenge

5.1 Comparison with Baselines on Ego4D v2

Top-5 mAP (%) on Ego4D v2 validation and test splits.

Method	Data Split	Noun (N)	N+V	N+TTC	Overall
FRCNN+SF [14]	Val	21.00	7.45	7.04	2.98
StillFast [2]	Val	20.26	10.37	7.16	3.96
GANOV2 (ours)	Val	20.52	10.42	7.28	3.99
FRCNN+SF [14]	Test	26.15	9.45	8.69	3.61
StillFast [2]	Test	25.06	13.29	9.14	5.12
GANOV2 (ours)	Test	25.67	13.60	9.02	5.16

Explanation.
Here GANOV2 is evaluated against FRCNN+SF and StillFast. On the validation set it offers a small but consistent improvement over StillFast across all metrics. On the test set, GANOV2 improves Noun and N+V slightly compared to StillFast while maintaining comparable N+TTC and Overall, showing that guided attention can refine predictions without redesigning the whole architecture.

5.2 Guided Fusion Layer Ablation

Ablation over which 3D CNN output layer uses Guided Attention Fusion, measured on Ego4D v2 validation (Top-5 mAP %).

Guided Fusion Layer(s)	Noun (N)	N+V	N+TTC	Overall
Layer 1	18.70	9.42	6.27	3.22
Layer 4	20.47	10.40	7.20	3.96
All layers	20.52	10.42	7.28	3.99

Explanation.

This ablation shows how the choice of fusion layer affects performance. Using guided attention only at early layers (Layer 1) underperforms. Applying it at a deeper layer (Layer 4) helps, and applying guided fusion across **all** layers gives the best scores on all metrics, indicating that multi-layer guidance better captures both low-level and high-level cues for next-active object anticipation.
